SINGAPORE UNIVERSITY OF SOCIAL SCIENCES

ICT330

Timed Online Assignment - July Semester 2022

Database Management Systems

Monday, 14 November 2022 01:00 pm - 03:00 pm

Time allowed: 2 hours

INSTRUCTIONS TO STUDENTS:

1. This Timed Online Assignment (TOA) contains **FOUR (4)** questions and comprises **SIX (6)** pages (including cover page).

2. You must answer **FOUR (4)** questions.

3. If you have any queries about a question, or believe there is an error in the question, briefly explain your understanding and assumptions about that question before attempting it.

4. You MUST **submit your answers via Canvas (similar to TMA submission) at the end time of this TOA** (as stated on this cover page). The 15 minutes grace period as shown on Canvas is strictly meant for technical issues encountered during submission. Thereafter, you will not be able to submit your answers and you will be considered as having withdrawn from the course. **No appeal will be allowed.**

5. Your submission should consist of only one file and **must not exceed 500MB** in size. The file must be a Microsoft Word file saved in .docx format. All answers are to be typed. Flowcharts and graphs may be scanned or photographed and embedded in the Word file provided it does not exceed the file size limit of 500MB.

6. To prevent plagiarism and collusion, your submission will be reviewed by Turnitin. The Turnitin report will only be made available to the marker and you will not be able to view it.

7. The University takes plagiarism and collusion seriously, and your Turnitin report will be examined thoroughly as part of the marking process.

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Answer all questions. (Total 100 marks)

Question 1

A small town has decided to implement contact tracing. Locations that have been marked for contact tracing are given a location identification number. Locations include buildings such as shopping centres and hospitals as well as units within a building.

Details of each visit include

- visitor id,

The visitor id is either a passport number or a national registration identity card number

- visitor contact

A visitor contact must be a local phone number

- date and time of entry
- date and time of exit
- location identification number

Whenever a case of infection is confirmed, the visits of the infected person that occur during the past 7 days of the confirmation date of infection are retrieved. Using the location identification number of these visits,

- address and building name can be extracted if the location is a building,
- unit number, the address and building name can be extracted if the location is a unit in a building.

Based on the duration of visits of the infected person, the visited locations may be subjected to disinfection. In addition, the visitors whose visits coincide with the retrieved visits of the infected person will be notified to observe their health for the next 14 days.

Construct a conceptual (ER) model from the statements of requirements to represent the data model for contact tracing, showing:

Question 1a

Entities, with entities name, relevant attributes and identifier,

Note to students: State your assumption(s) for only data requirements that are not specified.

(10 marks)

Question 1b

Relationships with maximum and minimum cardinalities and relationship name.

Note to students: State your assumption(s) for only data requirements that are not specified.

(10 marks)

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Question 2

To minimise the spread of infectious disease, an infected person is placed in one of several facilities for management. Facilities are graded based on the level of health care they provide to an infected person.

When an infected person is sufficiently well, he is placed in a facility with the lowest level of health care. Testing to detect whether a person is still infected happens only in facilities with the lowest level of health care. Tests are administered every three days. Once the person has 3 negative test results, the person has fully recovered and can be discharged.

It is possible that after a person has improved and testing starts, he can later deteriorate. Thus, an infected person may be moved back to a facility with the grade or level of health care he requires.

As part of infectious disease management, all addresses of the infected person are recorded e.g., addresses of home and hostel. Only one contact number, however, is necessary.

You are given the Case table:
Case (personId, personContact, personAddress, facilityId, facilityGrade, startStayDate, endStayDate, dateTested, testResult)

State your assumption(s) for only data requirements that are not specified.

Question 2a

Formulate and list the functional and multi-valued dependencies.

(7 marks)

Question 2b

Propose candidate key(s) for the table.

(2 marks)

Question 2c

Question 2c(i)
For each MVD in answer to part a), state whether it is subsumed.

(2 marks)

Question 2c(ii)
Normalise the table to BCNF and 4NF by applying the BCNF approach covered in the course text.
Show how you apply the steps to arrive at the tables, and explain whether each derived table at each iteration is already in BCNF and 4NF (if applicable).

(6 marks)

Question 2c(iii)
Assemble your BCNF and 4NF tables and constraints clearly by giving each table and the referential integrity constraint(s), if any, in the forms described below:

Tables in this form:
TABLE-NAME(col-name$_1$, col-name$_2$, ..., col-name$_n$)

- indicate (PK) against the primary key and
- indicate (FK) against the foreign keys, if any.

Referential integrity constraint(s), if any, **together with the child table** in this form:
TABLE-NAME$_x$.col-name$_{x1}$ ...col-name$_{xn}$ must exist in
TABLE-NAME$_y$.col-name$_{y1}$...col-name$_{yn}$

(8 marks)

Question 3

Given the following ERD in Figure Q3.

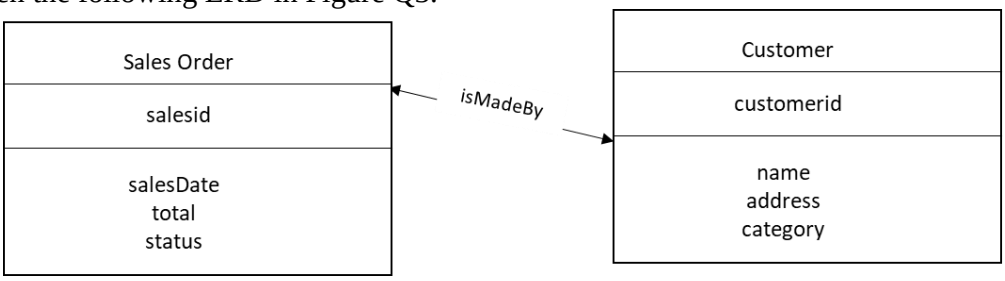

Figure Q3

Question 3a

Transform the ERD in Figure Q3 into a logical model.
Refer to the description of the columns for the column definition, and then write the appropriate SQL statements to implement the tables: SalesOrder and Customer, derived from the ERD in Figure Q3.

List your SQL statements in the order that they should be executed.

Description for columns in the tables:

- SalesOrder
  - all columns are required,
  - salesId is a surrogate key with value starting at 10000 and incremented by 1
  - salesDate is datetime, the default date is today,
  - total is decimal number with 2 digits for cents, and it ranges from $15 to $100
  - status is exactly 16 characters long, and is one of these values: under processing, waiting delivery, delivered, cancelled. The default value for status is under processing.

- Customer
  - all columns are required,
  - customerId is exactly 5 character long, and starts with two letters followed by 3 digits,
  - name is at most 25 characters long,
  - address is at most 40 characters long,
  - the columns name and address, combined, form an alternate key.
  - Category is a whole number from 1 to 5.

- Constraint
  C1: Minimum cardinality between Customer and SalesOrder for the isMadeBy relationship is 1:N, M:O with delete and update disallowed.

(10 marks)

Question 3b

Formulate SQL queries to display the required result sets:

Question 3b(i)
Display the details of sales orders that are cancelled this month. Show the later dates first.

(4 marks)

Question 3b(ii)
Display details of customers who have not cancelled any orders.

(4 marks)

Question 3b(iii)
Display details of customers who are in categories where there are more than 10 orders cancelled.

(7 marks)

Question 3c

Write a trigger to enforce the constraints:

- A customer who has cancelled 2 orders in the past 14 days cannot make an order.
- If one order has been cancelled in the past 14 days, the order cannot exceed the average total of all his past delivered orders. If there is no order previously delivered to the customer, then the average total of all his past delivered orders is 0 (zero).

State any assumptions you made in implementing the trigger.

(10 marks)

Question 4

Question 4a

Demonstrate the basic techniques of concurrency control and their application.

Consider a fictitious scenario where Mary and John have a joint bank account. John is at a shop and he is purchasing some expensive electronic goods. The system checks the account balance and as there is sufficient money, the system proceeds with the purchase. Meanwhile, Mary is in the midst of transferring money from her own account into the joint account, and this explains why there is now sufficient money in the joint account for John's purchase.

Question 4a(i)
Identify the level of isolation that allows the scenario to happen. Explain how you make the identification.

(3 marks)

Question 4a(ii)
Does this level of isolation pose a problem? If so, name the problem and explain the problem in terms of the scenario. If the level of isolation does not pose a problem, explain how the level of isolation correctly isolates John's transaction and Mary's transaction.

(3 marks)

Question 4a(iii)
Can a deadlock happen at this level of isolation? Explain either through the locking mechanism of this isolation level or through the use of an example.

(2 marks)

Question 4a(iv)

Can a lost update happen at this level of isolation? Explain either through the locking mechanism of this isolation level or through the use of an example.

(4 marks)

Question 4b
Suppose Mary's transaction is committed but John's transaction is aborted. Highlight and explain the differences in the database recovery procedure when handling the two transactions.

(4 marks)

Question 4c
Consider the Customer and SalesOrder tables in Q3. Describe **ONE (1)** permission to give to customers who make an online sales order. Describe **ONE (1)** permission not be given to customers.

(4 marks)

----- END OF PAPER -----